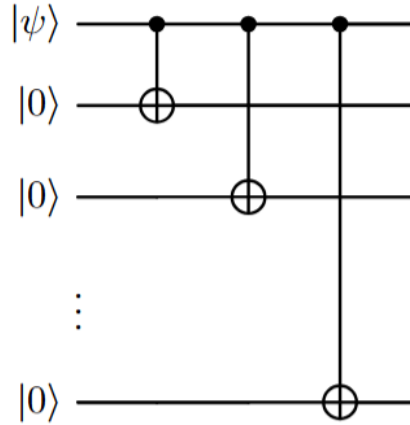


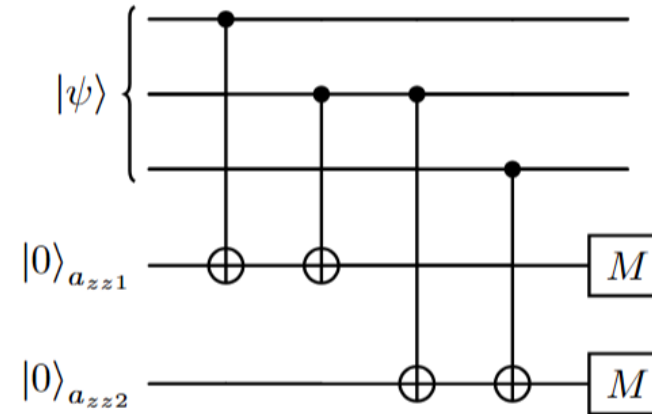
Correcting Bit Flip Errors with a Neural Network Decoder

Circuit Design



$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \rightarrow \alpha |000\dots 0\rangle + \beta |111\dots 1\rangle$$

- Encode arbitrary state across n qubits



- Use stabilizers to extract error

Design

- Noise created by bit-flip errors at gates
- Train neural network to label Pauli errors using stabilizer syndromes
- Associate labels with unitary corrective frame that maximizes logical fidelity

$$\rho \rightarrow U\rho U^\dagger$$

$$U_{X \rightarrow q_1} = X \otimes I \otimes I$$

Neural Network Decoder

Overview:

- Syndrome bits are extracted from circuit → training data
- Syndrome bits → Error labels → Corrective frame
- Neural network finds an optimal correction

Neural Network Decoder

1) Pre-Training

- Ideal circuit is injected with errors
- Syndrome bits associated with error
- Candidate map for frame is created

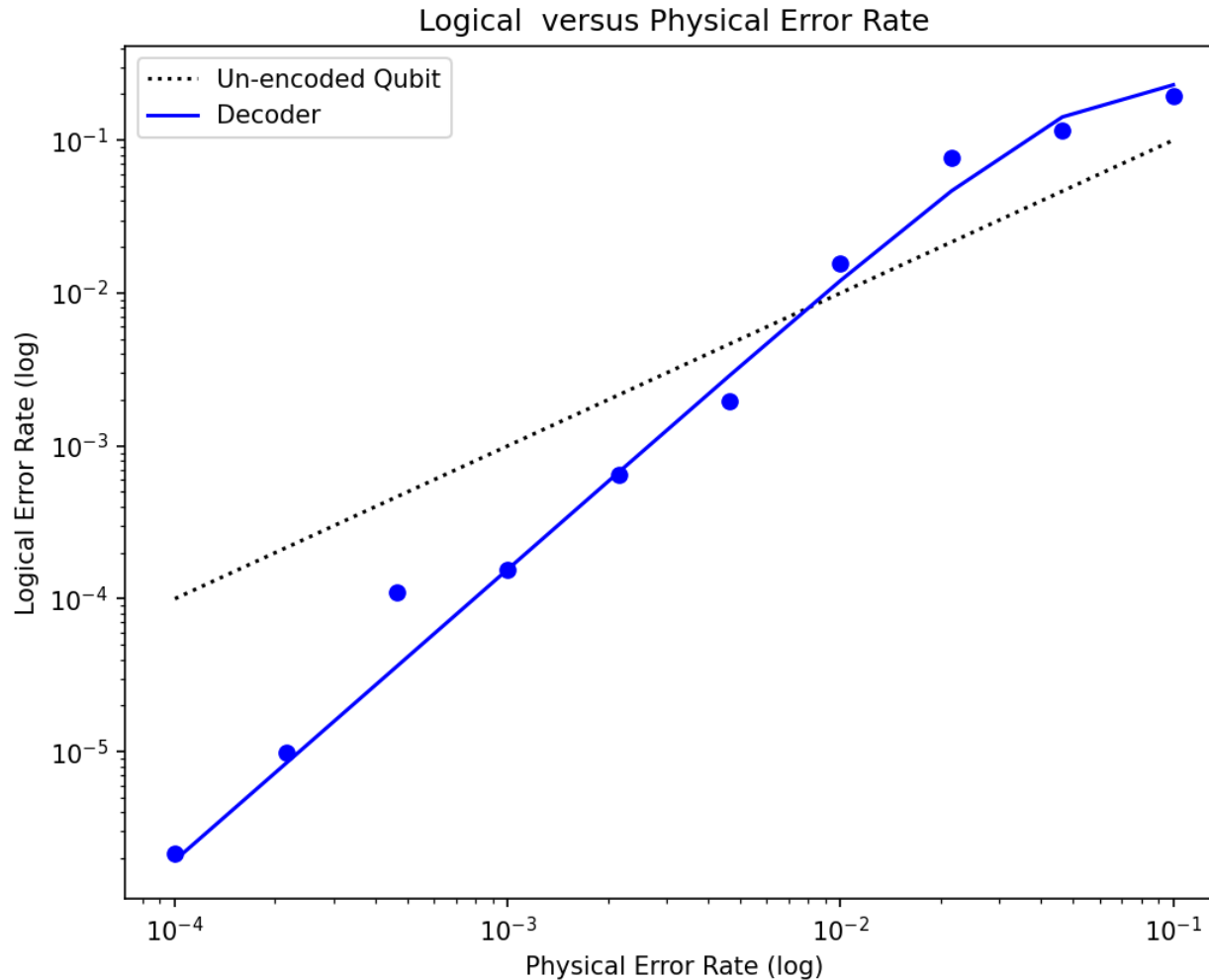
2) Fine-tuning

- Noisy circuit trains NN to find optimal frame

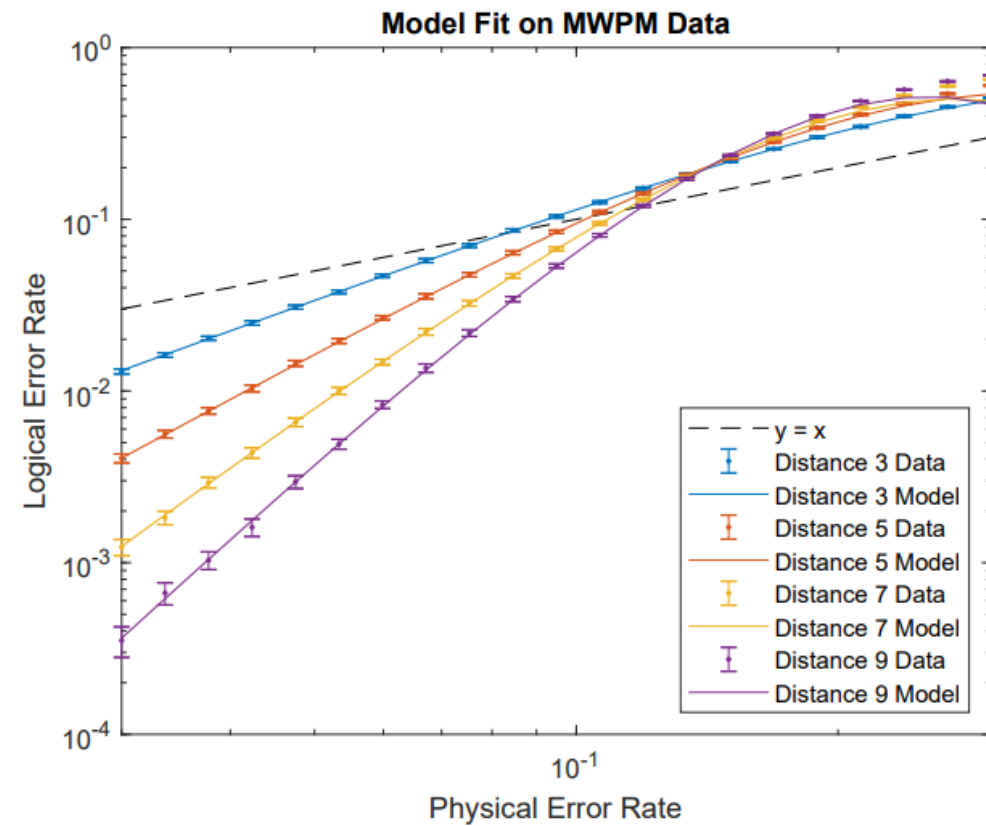
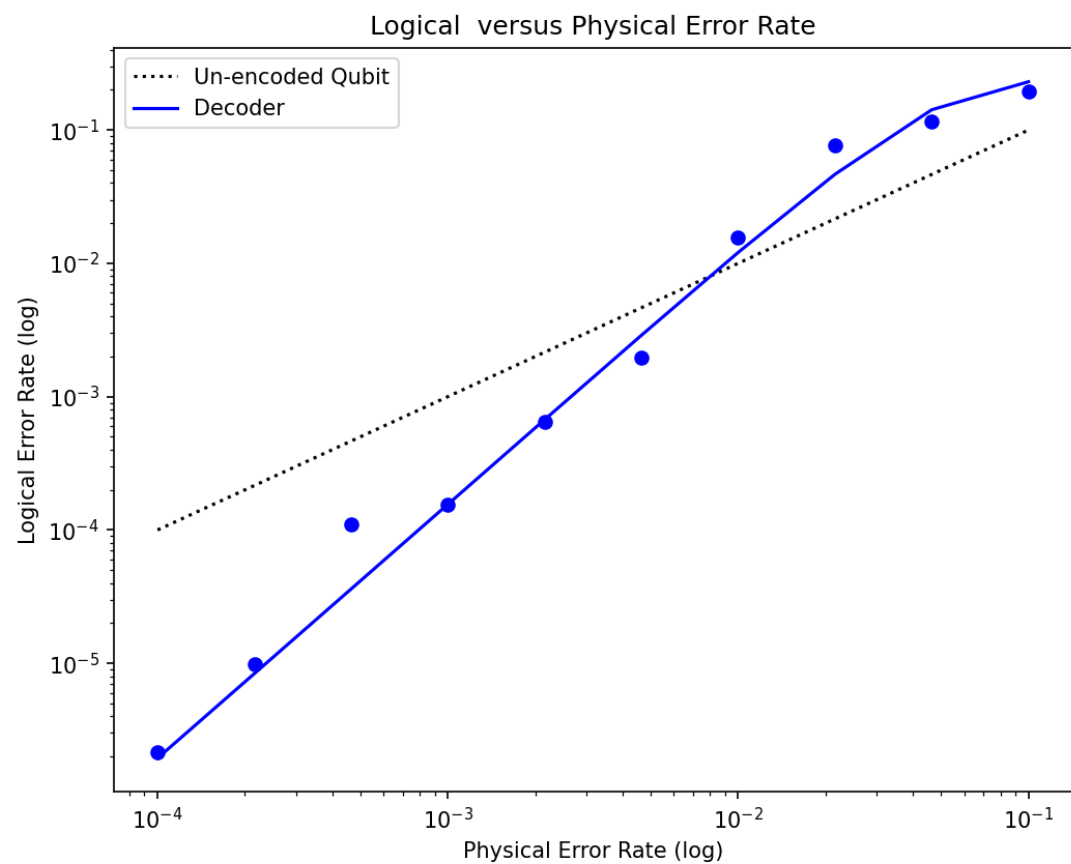
3) Simulation

- NN predicts correct frame on noisy circuit

Results



- Un-encoded qubit is modeled using $y = x$
- Decoder line is modeled using:
$$p_L = p_{th} \left(\frac{p_{phys}}{p_{th}} \right)^s (1 - c * p_{phys})$$
- $p_{th} = 0.008$, $s = 1.9$, $c = 3$



References

- [1] Overwater, Ramon, “Neural-Network Decoders for Quantum Error Correction”, arXiv, 2022.